HEMANTH KUMAR MUDADLA

SOFTWARE PROGRAMMER | AI ENGINEER | GENERATIVE AI | DATA & ENTERPRISE SYSTEMS

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PROFESSIONAL PROFILE

Computer Science professional with experience across enterprise software, artificial intelligence, data analytics, machine-learning workflows, and DevOps. Current work focuses on LLM-powered enterprise applications, RAG, semantic and hybrid search, NL2SQL, federated data access, unstructured-data processing, model recommendation, multilingual voice AI, and agentic AI. Strong hands-on foundation in Python, SQL, enterprise data platforms, APIs, validation, and production-oriented AI engineering.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, ETL, structured and unstructured data processing
Generative & Agentic AI: LLMs, RAG, Graph RAG, prompt engineering, agentic workflows, voice AI, multilingual AI, NL2SQL
AI / ML: TensorFlow, PyTorch, Hugging Face, Scikit-Learn, model evaluation and recommendation
Search & Retrieval: Semantic search, hybrid retrieval, vector databases, metadata filtering, citation-aware retrieval
Data Platforms: PostgreSQL, Trino, federated SQL, multi-database querying, metadata and data governance
Engineering: REST/API integration, Linux, Git, CI/CD, RBAC, data masking, validation and testing
Visualization: Tableau, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

Solix Technologies, Inc. | Software Programmer / AI Engineer **Apr 2025 - Present**
Lewisville, Texas, USA

- Develop enterprise AI capabilities using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Graph RAG, semantic retrieval, and prompt-driven workflows for structured and unstructured enterprise information.
- Design and enhance semantic-layer and Natural Language-to-SQL (NL2SQL) workflows that translate user questions into governed SQL and validate generated queries for correctness and safe execution.
- Implement Trino-based federated SQL execution across heterogeneous enterprise databases, simplifying multi-source data access while preserving RBAC, data masking, metadata, and governance controls.
- Build and validate unstructured deep-search workflows covering document parsing, preprocessing, chunking, sentence splitting, business-entity and metadata extraction, normalization, hybrid retrieval, filtering, and citation/document attribution.
- Developed a model-recommendation service/API using enterprise context inputs such as project, domain, subdomain, and task to return ranked top-N LLM recommendations for downstream AI workloads.
- Develop and improve a multilingual conversational voice AI sales agent integrating speech recognition, LLM responses, RAG-grounded product knowledge, text-to-speech, lead capture, email/phone validation, demo-booking flows, and conversation handling.
- Worked on multilingual voice-agent behavior across English, Hindi, Spanish, and additional supported languages, including language detection, response brevity, latency, voice/accent behavior, and customer-facing demo quality.
- Contribute to AI-agent and agentic workflow design, including orchestration concepts, enterprise use cases, tool/model selection, structured inputs and outputs, and integration with broader AI platform capabilities.
- Contribute to AI CMO / marketing strategy agent design, including the Marketing Priorities Agent for prioritization, portfolio management, strategic initiatives, investment scenarios, roadmap generation, risks, dependencies, KPIs, and structured outputs.
- Integrate and test PostgreSQL and enterprise data sources for AI/search workflows; use local database environments when needed to validate pipelines, data models, and end-to-end processing.
- Perform independent validation and troubleshooting of AI-generated SQL, retrieval results, API responses, search outputs, and JSON/report structures to improve accuracy, reliability, and production readiness.
- Collaborate with engineering, product, platform, QA, and stakeholder teams to translate business requirements into AI capabilities, demonstrate solutions, incorporate feedback, resolve integration issues, and improve latency and response quality.

OriginHubs | DevOps Engineer **Aug 2024 - Apr 2025**
Texas, USA

- Developed Python and Scikit-Learn data-processing pipelines supporting machine-learning workflows and repeatable data preparation.
- Engineered ETL workflows for structured and unstructured datasets and automated data-quality validation using Linux-based scripting.
- Supported cloud-native deployment and CI/CD automation for machine-learning systems and development workflows.

Teleperformance (Client: Google) | Data Analyst - Waymo Project **Aug 2019 - Feb 2022**
Hyderabad, India

- Analyzed large-scale sensor and telemetry data for the Waymo autonomous-driving program, supporting data-quality and safety-focused machine-learning workflows.
- Processed and prepared data using Python and SQL for downstream model training, analysis, and operational reporting.
- Built analytical reports and dashboards and performed root-cause analysis to identify data and system-quality issues.

EDUCATION

Master of Science in Computer Science | Governors State University, Illinois, USA | Aug 2022 - May 2024
Bachelor of Technology in Computer Science | Institute of Aeronautical Engineering, Hyderabad, India | Aug 2015 - Jun 2019

CERTIFICATIONS

IBM - Python for Data Science, AI & Development | Google AI Essentials | Great Learning - Project Management